Expt. 12 Polarization of light

Ⅰ. Purpose

(1) Get familiar with various methods to obtain the polarized light.

(2) Skillfully use instruments to create and verify polarized light.

(3) Understand the principles and effects of wave plates.

Ⅱ. Apparatus

One sodium lamp, one semiconductor laser, two polarizing plates, $\lambda/2$ wave plate, $\lambda/4$ wave plate, collecting lens, diaphragm, receiving screen, one for each item, one photoelectric receiver, an optical bench.

Ⅲ. Principle

1. Obtaining of polarized light

There are three methods used to obtain polarized light from natural light, they are:

(1) Utilize Brewster's Law to generate polarized light

$$n_{21} = \tan i_\beta \tag{12-1}$$

This incident angle i_β is called the Brewster angle, equation (12-1) is Brewster's Law. It is clear that the value of i_β has a close relation to the characteristics (refractive index) of the two media.

(2) Use the dichroism of a polarizing plate to generate linearly polarized light

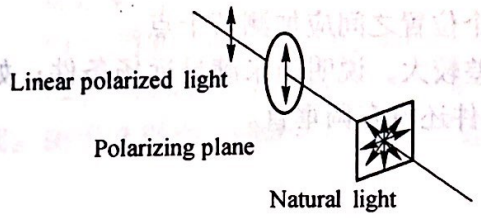


Figure 12-1 Creating Linear Polarization Using a Polarizing Plate

The characteristic of a material to selectively absorb one of the two perpendicularly polarized components in a natural light wave while hardly absorbing the other is called the dichroism of the material. The direction in which the light vector is allowed to pass through the polarizing plate without absorption is usually called the polarization direction of the polarizing plate (see Figure 12-1).

(3) Use the birefringence of a crystal to generate linearly polarized light

The phenomenon of birefringence occurs when natural light passes through an anisotropic crystal. The two beams of light generated by birefringence are linearly polarized. When the incident natural light is incident perpendicular to the crystal plane, one beam obeys the refraction law. This is called the Ordinary ray, or o ray. The other beam does not and it is called the Extraordinary ray or e ray. The polarization vector direction of the o ray is perpendicular to the principal section of the crys-

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tal (principal section is the plane that contains the optic axis of the crystal and the incident beam); the polarization vector direction of the e ray is in the principal section of the crystal.

2. Verification of polarized light

After natural light passes through the polarizing plate P, it becomes polarized light. If we place a Polarizing Plate A behind another Polarizing Plate P, the intensity of light which passing through A obeys Malus law:

$$I = I_0\cos^2\theta \qquad (12-2)$$

where, I_0 is the light intensity of the polarized light that gets through Polarizing Plate P, and θ is the angle between the plane of polarization of P and that of A shown in Figure 12-2.

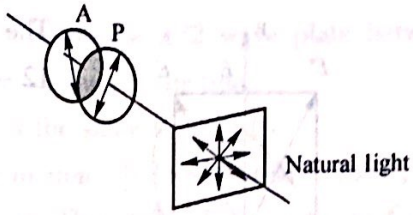


Figure 12-2 Verification of Malus Law

3. Wave plate, circularly polarized light and elliptically polarized light

(1) Wave plate

The so called "Wave Plate" is a slim plate cut from a single axis birefringent crystal. When passing through the wave plate, polarized light is divided into an o ray and an e ray. After passing through the wave plate there is a phase difference δ between the o ray and the e ray. Assume that the thickness of the crystal is d, then the optical path of the o ray is $n_o d$, the optical path of the e ray is $n_e d$ and the phase difference between the two beams of light is:

$$\delta = \frac{2\pi}{\lambda}(n_e - n_o)d \qquad (12-3)$$

Where λ is the wavelength of the incident light, n_o and n_e are the refractive indices of the crystal to o rays and e rays respectively. The phase difference between the o ray and the e ray after passing through the wave plate is:

$$\Delta = (n_e - n_o)d \qquad (12-4)$$

(2) Full wave plate, half wave plate and 1/4 wave plate

The thicker the wave plate, the greater the additional optical path difference and phase difference between the o ray and the e ray which emerge. When the thickness is exactly equal to a whole wavelength or odd multiple of the whole wavelength, the plate is called whole wave plate.

When the thickness is exactly equal to a half wavelength (or odd multiple of half wavelength), the plate is called a half wave plate or 1/2 wave plate.

When the thickness is exactly equal to 1/4 wavelength (or odd multiple of 1/4 wavelength); and the phase difference between the two beams of light is exactly equal to odd multiple of $\pi/2$, the plate is called a 1/4 wave plate.

(3) Linearly polarized light, circularly polarized light and elliptically polarized light

When a light wave travels along the z direction of, then:

$$E_x = E_e\sin(\tilde{\omega}t + \phi + \delta)$$

$$E_y = E_o\sin\tilde{\omega}t$$

where, E_e and E_o are the amplitudes of the e ray and o ray, ϕ is the relative phase difference between the e ray and the o ray in the incident polarized light. Their combined vibration equation is:

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$$\frac{x^2}{E_e^2}+\frac{y^2}{E_o^2}-\frac{2xy}{E_eE_o}\cos(\delta+\phi)=\sin^2(\delta+\phi) \tag{12-5}$$

① When $\delta+\phi=2k\pi$ $(k=0, \pm1, \pm2, \ldots)$, equation (12-5) changes to:

$$\left(\frac{x}{E_e}-\frac{y}{E_o}\right)^2=0, \text{i. e. } \frac{x}{E_e}=\frac{y}{E_o}$$

The result of this combination is shown as the solid line E in Figure 12-3.

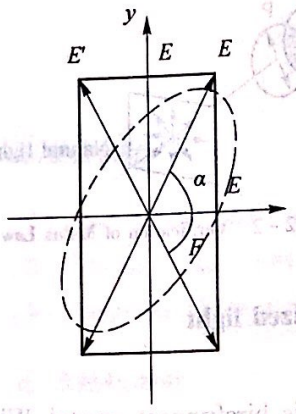


Figure 12-3 Linearly Polarized Light and Elliptically Polarized Light

When $\delta+\phi=(2k+1)\pi$, equation (12-5) changes to:

$$\left(\frac{x}{E_e}+\frac{y}{E_o}\right)^2=0, \text{i. e. } \frac{x}{E_e}=-\frac{y}{E_o}$$

The emergent light is still linearly polarized light in both cases, as shown by the solid lines E′ in Figure 12-3.

② When $\delta+\phi=(2k+1)\frac{\pi}{2}$, equation (12-5) changes to:

$$\frac{x^2}{E_e^2}+\frac{y^2}{E_o^2}=1 \tag{12-6}$$

The combined vibration gives elliptically polarized light, as shown by the dashed line in Figure 12-3.

Ⅳ. Experimental

1. Verify Malus law

(1) Adjust the apparatus to make sure that the light source and components have a common axis, as shown in Figure 12-4. Turn the polarizing plate P, and observe the change in intensity of the transmitted polarized light on the white screen. Record the changing results, and explain the reason for such a change.

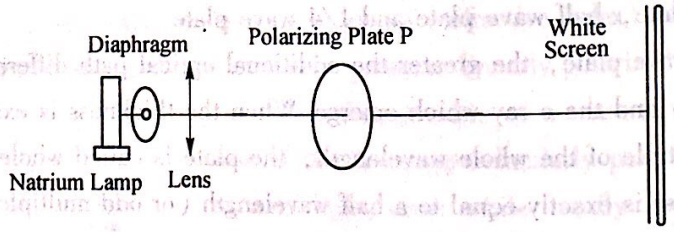


Figure 12-4 Generation of Linear Polarization Poalrizer P

(2) Insert the polarizing plate A behind P as the analyzer, then turn A and observe the change in light intensity on the white screen. Explain reasons for the changes.

(3) Verify Malus law. First align the polarizer P and the analyzer A to give light extinction as measured by the photoelectric receiver. Obtain and record the light intensity as A is rotated from 0° ~ 180° in intervals of 10°. Draw the I versus $\cos^2\theta$ relationship curve, and give a brief explanation and discussion of the curve.

2. Observe the phenomenon of polarized light passing through a $\lambda/2$ wave plate

The ray diagram is shown in Figure 12-5.

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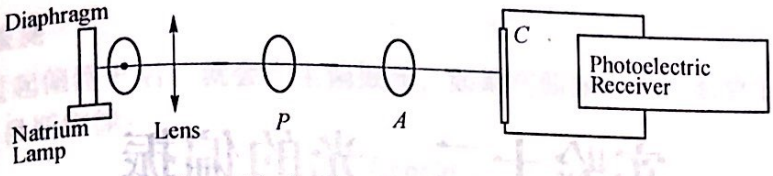


Figure 12 – 5 Experiment Light Path Diagram

(1) Adjust P and A, to give light extinction, and then insert the $\lambda/2$ wave plate between them. Turn the half wave plate through 360° noting the numbers of light extinction.

(2) Fix P and $\lambda/2$ wave plate, with the other procedures the same to Step (1).

(3) Align the polarizer P and analyzer A to give light extinction; holding P and A fixed, insert the $\lambda/2$ wave plate and turn it to give light extinction again. Then turn P clockwise to f until light is transmitted and then turn A anticlockwise to an angle α, to again achieve the light extinction position. Observe and record the data and explain the physical reasons based.

3. Observe the phenomenon of polarized light passing through a $\lambda/4$ wave plate

(1) With procedures the same to Step 2 (2), replace the $\lambda/2$ wave plate with the $\lambda/4$ plate, and put the system into a state of light extinction again. Turn P to an angle 0°, then turn angle A from angle 0° to 360° in intervals of 30°, recording the light intensity I. Repeat the above steps and observe the light intensity with A going from 0° to 360° at angles of 15°, 45°, 60°, 75° and 90° etc.. Draw separate $I-\theta$ relationship curves for the two sequences of angles given.

(2) From the $I-\theta$ relationship curves, find out the light intensity ratio of the transmitted light I_{min}/I_{max}, and explain when the transmitted light is linearly polarized light, circularly polarized light or elliptically polarized light.

***4. Observe and measure the polarization characteristics of a semiconductor laser (optional)**

Select a semiconductor laser as the light source and determine the polarization characteristics of the semiconductor laser by experiment. You should design the experiment scheme yourselves and from the experimental results determine what kind of polarized light is produced by the laser.

Notes:

(1) Must not touch the surface of the polarizing plates or the wave plates with your hands.

(2) Switch on the sodium lamp before the experiment for preheating. You must not switch off the sodium lamp during the experiment.

(3) Avoid stray light entering the photoelectric receiver as much as possible during the experiment.

V. Questions

1. Linearly polarized light is incident on a wave plate.

Determine whether the light is transmitted or absorbed under the following conditions:

(a) Light is polarized perpendicular to the optic axis of a $\lambda/2$ wave plate.

(b) Light is polarized parallel to the optic axis of a $\lambda/2$ wave.

Repeat (a) and (b) for a $\lambda/4$ wave plate.

Explain all your answers.

2. How could you make a 1/2 wave plate and a full wave plate from several 1/4 wave plates?

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实验十二　光的偏振

一、实验目的

（1）熟悉各种偏振光的获得与检验的方法。

（2）熟练使用产生与检验偏振光的器件。

（3）了解波片的原理和作用。

二、实验仪器

钠光灯一台，半导体激光器一台，偏振片两片，$\lambda/2$ 片，$\lambda/4$ 片，聚光透镜，光阑，接收屏各一个，光电接收器一台，光具座一套。

三、实验原理

按振动形式，光可分为线偏振光、圆偏振光、椭圆偏振光、部分偏振光和自然光。线偏振光是其中最基本、应用上十分重要的一种。

1. 偏振光的获得

从自然光中获得偏振光的方法有三种，分别是

（1）利用布儒斯特定律产生偏振光

当自然光以某一个特定的角度 i_β 入射到界面上时，这时反射光中会只含有一个振动方向，另一个与它垂直的振动方向的分量完全没有，这种光通常称为线偏振光，反射光的振动方向垂直于入射面。这时透射光仍为部分偏振光。这一特殊的角度 i_β 满足下列关系：

$$n_{21} = \tan i_\beta \tag{12-1}$$

此入射角 i_β 称为布儒斯特角，式（12-1）为布儒斯特定律。很明显，式（12-1）中 i_β 的大小与两个介质的特性（折射率）有密切关系。

（2）利用偏振片的二向色性产生线偏振光

物质有选择地吸收光波中两个垂直分量之一，而对另一个振动分量基本不吸收的性质称为物质的二向色性。偏振片中允许光矢量无衰减地透过的方向，常称为偏振片的偏振化方向，如图 12-1 所示。

（3）晶体的双折射性产生线偏振光

自然光通过各向异性的晶体时会发生双折射现象，双折射产生的两束光为线偏振光。当入射光垂直于晶体表面入射时，其中一束光总是遵守折射定律，这束光称为寻常（Ordinary）光，也称为 o 光，而另一束光不遵守折射定律，被称为非寻常（Extraordinary）光，或称 e 光。o 光光矢量的振动方向垂直于自己的主截面（主截面是包含晶体光轴和入射光线的平面）；e 光光矢量振动方向在自己的主截面内。

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2. 偏振光的检测

当自然光通过起偏器 P 后，就会产生偏振光，如果在偏振片 P 后面放上偏振片 A，透过 A 的光强 I 满足马吕斯定律：

$$I = I_0\cos^2\theta \tag{12-2}$$

式中：I_0 是透过 P 的偏振光的光强，θ 是透过 P 和 A 偏振光的夹角，如图 12-2 所示。

3. 波片、圆偏振光和椭圆偏振光

(1) 波片

所谓“波片”，是从单轴双折射晶体上切下来的一个薄片，偏振光通过波片时，被分解成 o 光和 e 光，透过波片后，这两束光之间就产生了一个位相差 δ，假设晶片的厚度为 d，则 o 光的光程为 $n_o d$，e 光的光程差为 $n_e d$，这两束光的位相差为：

$$\delta = \frac{2\pi}{\lambda}(n_e - n_o)d \tag{12-3}$$

式 (12-3) 中 λ 为入射光波长，n_o、n_e 分别为晶体对 o 光和 e 光的折射率，透过波片后的 o 光和 e 光的光程差为：

$$\Delta = (n_e - n_o)d \tag{12-4}$$

(2) 全波片、半波片和 1/4 波片

波片的厚度愈厚，光线通过波片后，透射出来的 o 光与 e 光之间的额外光程差或位相差就愈大。当 Δ 正好等于入射光的波长（或波长的整数倍）的波片，称为全波片。

正好等于半个波长（或半个波长的奇数倍），称为半波片或 1/2 波片。

当正好等于 1/4 个波长（或 1/4 波长的奇数倍）；或者这两束光之间的位相差正好是 $\pi/2$ 的奇数倍，称这种波片为 1/4 波片。

(3) 线偏振光、圆偏振光和椭圆偏振光

当光波沿 z 方向传播时，有：

$$E_x = E_e\sin(\tilde{\omega}t + \phi + \delta)$$

$$E_y = E_o\sin\tilde{\omega}t$$

式中：E_e、E_o 为 e 光和 o 光的振幅，ϕ 为入射偏振光中 o 光与 e 光间的相对位相差。它们的合振动方程为：

$$\frac{x^2}{E_e^2} + \frac{y^2}{E_o^2} - \frac{2xy}{E_eE_o}\cos(\delta+\phi) = \sin^2(\delta+\phi) \tag{12-5}$$

这是一个椭圆方程，椭圆的形状取决于 $(\delta+\phi)$ 与 $\frac{E_o}{E_e}$ 值，即与晶片产生的额外位相差和入射偏振光的振动方向 OE 与晶片的光轴方向 OX 夹角 α 有关。其中，$E_e = E\cos\alpha$，$E_o = E\sin\alpha$。

① 当 $\delta+\phi = 2k\pi$ ($k=0, \pm1, \pm2, \cdots$) 时，式 (12-5) 变为：

$$\left(\frac{x}{E_e} - \frac{y}{E_o}\right)^2 = 0, \text{即} \frac{x}{E_e} = \frac{y}{E_o}$$

这就说明合成结果为线振动，如式 (12-3) 中实线 E 所示。

当 $\delta+\phi = (2k+1)\pi$ 时，式 (12-5) 变为：

$$\left(\frac{x}{E_e} + \frac{y}{E_o}\right)^2 = 0, \text{即} \frac{x}{E_e} = -\frac{y}{E_o}$$

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出射光仍为线偏振光，如图 12－3 中的 E' 实线所示所示。

② 当 $\delta+\phi=(2k+1)\frac{\pi}{2}$ 时，式（12－5）变为：

$$\frac{x^2}{E_o^2}+\frac{y^2}{E_o^2}=1 \tag{12-6}$$

合成振动为椭圆偏振光，如图 12－3 中的椭圆虚线所示。利用 1/4 波片可以将椭圆偏振光转变为圆偏振光。

四、实验内容及数据处理

1. 验证马吕斯定律

（1）调节光源与各元件共轴如图 12－4 所示，转动偏振片 P，用白屏来观察光束经 P 后的线偏振光的光强变化。记录变化结果，说明变化原因。

（2）在 P 后插入偏振片 A 作为检偏器，然后转动 A，由白屏观察光强变化。说明变化原因。

（3）验证马吕斯定律。先把起偏器 P 和检偏器 A 处于消光状态，利用光电接收器获取 A 在转角范围 0°～180°之间，记录每间隔 10°所对应的光强 I，并作出 $I-\cos^2\theta$ 关系曲线，并对曲线作扼要的解释和讨论。

2. 观察线偏振光通过 λ/2 片的现象

光路图如图 12－5 所示。

（1）调节 P 和 A，使它们处于消光位置，然后插入 λ/2 波片，再转动 360°，观察消光次数，并解释之。

（2）固定 P 与 λ/2 波片，其余步骤同（1）。

（3）将转动起偏器 P 和检偏器 A 处于消光位置，此时 P 与 A 不动，插入 λ/2 波片，使其转至消光位置，然后顺时针方向转动 P（每隔 15°）到 ϕ，此时消光状态消失，再令 A 沿逆时针方向旋转角度 α，使 A 达到消光位置，按下表观测数据并解释物理原因。

3. 观察偏振光通过 λ/4 波片后的现象

（1）步骤同 2（2），用 λ/4 波片代替 λ/2，并使系统再次达到消光状态。将 P 转动 0°，再将 A 从 0°角开始每隔 30°转动一次并记录光强 I，直到增加到 360°为止。重复以上步骤，接着观测 15°、45°、60°、75°、90°所对应的 A 从 0°～360°的光强，并分别作出 $I-\theta$ 关系曲线。

（2）从关系曲线找出透射光的光强比 I_{min}/I_{max}，并说明何时透射光为线偏振光、圆偏振光或椭圆偏振光。

*4. 观察测量半导体激光器的偏振特性（选做）

选用半导体激光器做光源，通过实验来确定半导体激光器的偏振特性，请同学们自己设计实验方案、实验表格，并通过实验结果进一步分析半导体激光器发出的光，是属于哪一种偏振光。

注意：

（1）禁止用手触摸偏振片和波片的表面。

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(2) 实验前先将钠光灯电源打开预热，实验中不得随意将钠光灯关闭。

(3) 实验时尽量避免杂光进入光电接收器。

五、思考题

1. 分别写出，当入射偏振光的振动方向与 $\lambda/2$ 波片的光轴垂直时和平行时，穿过晶片后的光斑是明的还是暗的？所用的波片取用 $\lambda/4$ 波片后，上述结果是否相同？为什么？

2. 如何将几个相同的 1/4 波片组成 1/2 波片和全波片？

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实验十　偏振光的研究

1. 验证马吕斯定律

(1)先将钠光灯打开,预热几分钟,调节光源位置,然后把光阑放在钠光灯前,在导轨上放入聚光透镜,调节透镜焦距,使光阑通过透镜正好成像在接收屏 H 上,再将起偏器 P 插在透镜与接受屏中间,调节 P 的高低位置,使光束与导轨平行,光束直接通过起偏器 P 的中心,该光束通过 P 后成为线偏振光,以入射方向为轴转动起偏器 P,用白屏 H 来观察光束经起偏器 P 后的光强变化。说明变化原因。

(2)固定起偏器 P,在 P 后插入偏振片 A 作为检偏器,A 的高度要与 P 一样,从 P 发出的光束通过 A 的中心,然后转动检偏器 A,用白屏 H 观察光强变化,记录光强变化结果,说明变化原因。

(3)验证马吕斯定律

检偏器转角 θ	光强 $I(V)$	$\cos^2\theta$	检偏器转角 θ	光强 $I(V)$	$\cos^2\theta$
0°			100°		
10°			110°		
20°			120°		
30°			130°		
40°			140°		
50°			150°		
60°			160°		
70°			170°		
80°			180°		
90°					

用直角坐标纸作出光强 I 与检偏器 A 转角 θ 的 $\cos^2\theta$ 值关系曲线,并对曲线作扼要的解释和讨论。

2. 观察线偏振光通过 $\lambda/2$ 片的现象

(1)转动起偏器 P 和检偏器 A,使它们的振动方向处于消光位置,然后将 $\lambda/2$ 波片插放在 P 与 A 之间,调节其高度让 P、A 相互平行,并使从 P 发出的偏振光正好通过 $\lambda/2$ 波片中心,再转动 $\lambda/2$ 波片一圈,即转 360°,观察消光次数,并解释物理原因。

(2)转动起偏器 P 和检偏器 A,使它们的振动方向处于消光位置,然后将 $\lambda/2$ 波片插放在 P 与 A 之间,调节其高度让 P、A 相互平行,并使从 P 发出的偏振光正好通过 $\lambda/2$ 波片中心,将 $\lambda/2$ 波片的光轴任意取向后,固定 P 与 $\lambda/2$ 波片,转动检偏器 A 一圈,观察消光次数,并解释物理原因。

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(3)将转动起偏器 P 和检偏器 A 处于消光位置,此时 P 与 A 不动,插入 $\lambda/2$ 波片,使其转至消光位置,固定好 $\lambda/2$ 波片,然后顺时针方向转动起偏器 P 到 φ,此时消光状态消失,再逆时针方向旋转检偏器 A,使 A 重新达到消光状态,记录检偏器 A 转动的角度 α。

φ	15°	30°	45°	60°	75°	90°
α						

观察 φ 与 α 的关系,并解释物理原因。

3. 观察偏振光通过 $\lambda/4$ 波片后的现象

I/V P / A	0°	15°	30°	45°	60°	75°	90°
0°							
30°							
60°							
90°							
120°							
150°							
180°							
210°							
240°							
270°							
300°							
330°							
360°							

(2)从上表中分别将起偏器 P 所转的角度 φ 与检偏器 A 相对应角度 α 中的最大光强 I_{max} 和最小光强 I_{min} 找出来,分析透射光的光强比 I_{min}/I_{max},并说明何时透射光为线偏振光、圆偏振光或椭圆偏振光。

φ	α 转动 360° 光强变化情况	光强变化		光强比	$\lambda/4$ 波片射出的偏振态
		I_{min}	I_{max}	I_{min}/I_{max}	
0°					
15°					
30°					
45°					
60°					
75°					
90°					

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